

The Taming of the Bias: Measuring and Mitigating Category-Aware Gender Bias in Recommendations

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Recommenders are a vital part of our daily lives. We rely on them for tasks such as discovering new music and connecting job seekers with relevant opportunities. Given their crucial role, we must ensure recommendations are free from societal stereotypes. Therefore, evaluating and mitigating biases in them is essential. Previous work used to evaluate the fairness of recommended items fails to capture certain nuances, by mostly focusing on comparing performance metrics for different sensitive groups. In this paper, we introduce a set of comprehensive metrics for evaluating gender bias in recommendations. We highlight the importance of evaluating fairness at a granular level, which can be achieved with our set of metrics, using item categories, like genres for movies. Furthermore, we show that adding a category-aware fairness metric as a regularization term to the main recommendation loss during training can help effectively minimize bias in the models' output. We experiment on three real-world datasets, using five baseline models, to show the effectiveness of our metrics in evaluating gender bias. Additionally, we use two of our metrics as separate regularization terms, to demonstrate how incorporating them significantly reduces the bias in recommendations for different categories without substantial degradation in overall recommendation performance. The link to the code and datasets are provided here: [GitHub repository](#)

CCS Concepts: • **Computing methodologies** → **Artificial intelligence**; • **Information systems** → **Personalization**; **Recommender systems**.

Additional Key Words and Phrases: Fairness Metrics, Bias in Recommendations, Societal Stereotypes, Recommender System, Gender Bias

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1 Introduction

Recommender Systems (RS) help personalize item selections for users by providing suggestions based on their preferences and behaviors. These systems play a significant role in shaping the decision-making process, as they are used across different platforms like e-commerce sites, social media, and news organizations, influencing the products and information we are being exposed to. On such online platforms, recommender systems help enhance user engagement by personalizing and diversifying content. This helps users avoid information overload and helps them focus on options that reflect their past interactions. Despite these benefits, recommendation systems also

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raise concerns about the potential bias that can manifest in them. For instance, studies demonstrate that in some recommendation systems, changing just the gender of the candidate in a job search (while keeping the qualifications the same), can affect access to higher-paying job opportunities [19, 44]. RS algorithms are usually evaluated using measures such as RMSE (Root Mean Squared Error), NDCG (Normalized Discounted Cumulative Gain), precision, and diversity [75]. When only these metrics are considered, then the recommended items are deemed good when they match user preferences. These metrics can help evaluate the performance of the items recommended, but in recent years, there is a growing emphasis on evaluating and ensuring fairness as well [1, 2, 15, 24, 32, 56, 67].

For example, the authors in [21] present a fairness evaluation metric that is adaptable to different fairness notions like user-centric or item-centric. Despite significant progress in this field in recent years, there is a notable gap in developing robust methods for quantifying consumer-side bias accurately. Many of the metrics suffer from several limitations: (i) they over-simplify the notion of fairness, (ii) they fail to consider the ranked nature of recommended items, and (iii) they rely solely on a single type of fairness metric. A widely used approach to evaluating consumer-side fairness in recommendations, involves an adaptation of *Equal Opportunity* (refer to Section: 2.2), which seeks to equalize performance metrics (such as recall) or utility scores across user groups, such as male vs. female users [56, 65, 66, 68, 77, 91].

However, this approach is relatively straightforward, it can obscure disparities in recommendations across different item categories. To illustrate such disparities in recommendations, we refer to Figure 1, which shows, for several recommendation algorithms, an analysis of the proportion of action and romance movies among the top 10 recommendations for male and female user groups, together with the corresponding Precision@10 values for each group. Three key observations emerge here: (i) across all models, there is a clear imbalance in the proportion of romance and action movies recommended, as romance movies tend to be recommended more frequently to female users, while action movies are more commonly recommended to male users; (ii) precision@10 values for both male and female users are comparable across all models, suggesting that, according to this metric, the models appear to perform equally well for both groups; and (iii) the magnitude of bias varies across different recommendation models. These findings indicate that simply comparing performance metrics across sensitive attributes is insufficient to assess fairness in recommendation systems. Instead, more granular metrics are required to accurately quantify bias and ensure fair recommendations across different consumer groups. To underscore the significance of granular evaluation, we provide a brief overview here with a more detailed example presented in Section 3.1. Bias can arise in recommender systems due to stereotypical interaction patterns in the training data, leading to biased recommendations based on sensitive attributes. Such biases can confine users to filter bubbles, with a focus on stereotypical categories or narrowly defined preferences. Overexposure to specific types of content may have unintended consequences; for example, if a young male user is predominantly recommended action movies that portray violence in a glorified and consequence-free way, then it could contribute to adverse effects, including the desensitization of violence. While this example is illustrative, the potential consequences are far more severe in sensitive domains like news or job recommendations, where such biases can lead to significant societal harm.

In this paper, we aim to address the limitations of current fairness assessment metrics by designing a set of evaluation measures designed to quantify gender bias in recommendation models. Our objective is to ensure that they are diverse to capture subtle yet meaningful forms of bias that might not be immediately obvious. We propose to do this by incorporating item categories and rank (where applicable). Furthermore, we utilize two of these metrics in the optimization objective as part of the loss function, resulting in improved fairness of recommendations across different

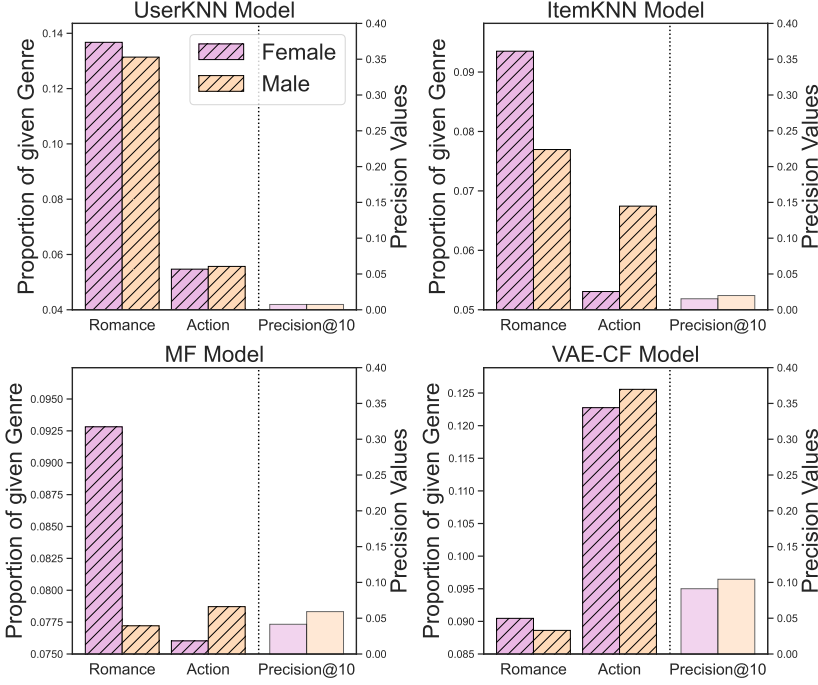


Fig. 1. Comparison of action and romance movie recommendations among male and female users across four recommendation algorithms, along with corresponding Precision@10 values. The graphs highlight disparities in genre recommendations by gender, with romance movies being more frequently suggested to female users and action movies to male users, despite similar Precision@10 metrics. Details of how this was generated is mentioned in Section A.1.

user groups within each category. This demonstrates that our metrics are effective in addressing fairness concerns but also proves their practical utility for quantifying bias. In addition, we apply our metrics to evaluate a range of recommendation models, using three real-world datasets. The contribution of this paper is summarized as follows:

- We propose a set of evaluation metrics that jointly account for item categories and rankings of the recommendations made, in turn providing a more nuanced evaluation.
- We apply the proposed metrics to analyze gender bias in different recommendation algorithms.
- We employ two in-processing optimization strategies to promote fairness in recommendation systems, not only in a general sense but in a category-aware sense.

We would like to mention that this paper is an extension of our conference paper accepted at the Web Conference (WWW) 2025 [52]. While the conference paper introduced a category-aware fair framework for evaluating and mitigating gender bias in recommendation systems using one regularization strategy, this version significantly expands both the methodology and experimental sections. Specifically, we introduce a second regularization term and conduct a comprehensive empirical comparison between the two regularization techniques. Additionally, we broaden the experimental study by evaluating our framework on two additional state-of-the-art recommendation models (LightGCN [42] and XSimGL [94]). The experiments were repeated using multiple random seeds to account for variability introduced by any stochasticity. Moreover, we update the related

work and other relevant sections, and improve the visualizations of results for clarity and interpretability. Collectively, these additions substantially strengthen the methodological, empirical and experimental contributions of the original conference paper.

2 Related work

We organize and discuss the relevant work existing in this field in the subsections below. Although a substantial body of research has addressed fairness and bias in recommender systems from a broad perspective [15, 81], this review primarily focuses on consumer-side fairness, specifically gender-related discrimination. The following section therefore surveys existing work on gender bias in recommender systems.

2.1 Gender Fairness in RS

Gender bias in recommendation systems can manifest as systematic discrepancies in how algorithms recommend items to users of different genders. Such bias can have a wide-ranging set of implications, including unequal representations of items across gender, reinforcement of stereotypes (e.g., male-dominated job roles recommended more frequently to male users), and reduced personalization, where recommendations made to users don't align with their actual preferences. Quantifying and mitigating gender bias in recommendation systems is an active area of research. Melchiorre et al. [66] examine the effects of resampling (a commonly used debiasing technique) on recommendation systems, finding that it slightly reduces gender bias, with a slight decrease in performance. Additionally, prior work [33, 37, 53, 89, 96] proposes ways to re-rank items to offer a balanced solution that caters to group fairness (for gender and other sensitive attributes) while still respecting individual user preferences. For instance, [33] presents re-ranking approaches for job candidate recommendations that are designed to ensure a specified distribution of sensitive attributes among the highest-ranked candidates. Graph-based recommender systems have increasingly incorporated fairness-aware techniques to mitigate gender bias by generating counterfactual interaction graphs [16], integrating fairness constraints into federated graph neural networks [4], and leveraging graph augmentation to reduce intersectional unfairness while preserving recommendation performance [8].

Historical data that reflect stereotypical viewing preferences can intensify certain biases further when used to train traditional recommendation models. For instance, male users exhibit a preference towards *action* movies, which is amplified by recommendation algorithms like UserKNN [78]. The authors in [3] propose a framework for fairly predicting the quality of TedTalk by leveraging causal models and counterfactual reasoning to mitigate gender and racial bias. Another line of work explores adversarial fairness, where the recommender system is trained not only to make accurate predictions but also to make it difficult for the adversary to guess the sensitive attribute [62, 71]. Boratto et al. [6] uses counterfactual augmentation of user-item graphs to reduce consumer unfairness, whereas our method instead regularises the recommendation objective using category-aware fairness signals and does not require graph editing or counterfactual edge generation. Recent advancements in this area, have led to the development of fairness-aware recommendation models, with particular focus on gender bias [9, 11, 53, 83, 88, 92, 98, 99].

2.2 Evaluating Gender Bias

In the evaluation of gender bias, the most commonly adopted fairness notions are *Demographic Parity* and *Equal Opportunity*, which are both related to group fairness. In this setting, fairness pertains to equitable treatment across different groups (which can be measured in classification and recommendation tasks). Group fairness aims to ensure that the predicted outcomes $\hat{Y}$ of a model are independent of the sensitive attributes like gender S . Under demographic parity, each

sensitive group (like male and female) should receive positive predictions at equal rates. For binary classification, this requirement can be formalized as:

$$P(\hat{Y} = 1|S = 1) = P(\hat{Y} = 1|S = 0)$$

This formulation implies that the probability of a positive outcomes is identical across both genders, where 0 may represent male and 1 may represent female or vice versa. Equal Opportunity, in contrast, is satisfied when a model has equal true positive rates across different demographic groups [40]. This concept can be formally defined as:

$$P(\hat{Y} = 1|S = 1, Y = 1) = P(\hat{Y} = 1|S = 0, Y = 1)$$

where Y represents the true outcome.

Beyond these two widely used methods to quantify fairness, some additional concepts (as discussed in [51]) used include Equalized Odds [40], Balance for Negative Class [79], Balance for Positive Class [79], Intersectional Fairness [36], Equal Calibration [18] and Causal-based notions [3, 55].

2.3 Evaluating Consumer-Side Fairness

In recommendation systems, fairness is widely recognized as a multi-sided concept and is categorized into three groups: consumers (C-fairness) [21, 23, 34, 37, 39, 66, 80, 86] which concerns the impact of recommendations on protected user groups, providers (P-fairness) which focuses on ensuring fair exposure for providers/sellers on a platform and both (CP-fairness) [13]. In this work we concentrate on consumer-side fairness as we want to ensure that recommendations made by models are not biased against a certain gender. Prior studies have demonstrated how recommendations can differ unfairly across sensitive attributes of users like gender, age, race, etc. [38, 49, 81]. Consequently, evaluating bias in recommendation systems, prior to deployment, is critical for preventing the reinforcement of stereotypes and limiting diverse content. When assessing consumer-side bias, a common approach is to adopt the notion of equality of opportunity and examine the differences in metrics such as recall, precision and/or NDCG [30, 56, 66]. Other works leverage causal-based fairness notions [58, 83] and demographic parity [9, 26, 33]. Unlike these metrics, which equate fairness strictly with equality, Deldjoo et al [21] argue that fairness may involve unequal treatment in certain contexts, like paid users should be provided better recommendations when compared to free users. They propose a flexible set of metrics that can take this disparity into account and then measure fairness accordingly. Another notable perspective is the concept of envy-free fairness, which is achieved when no user significantly prefers another user's recommendations than their own [22]. The concept of ϵ -fairness constrains the outcome from recommendation models across protected groups by a threshold to limit disparities for binary sensitive attributes [29, 45].

While these metrics can capture aspects of consumer-side bias, they can exhibit limitations that restrict their ability to fully characterize the unfairness in recommended items. Current metrics used to quantify consumer-side bias in recommendation systems suffer from several limitations. These include: (i) over-simplifying the notion of fairness in RS, which encompasses various techniques like collaborative and content-based filtering and hybrid methods. Such simplified definitions may fail to capture disparities across different items characteristics, for example, genres, when considering movie recommendations. Several works have used such metrics to quantify bias and are listed in Table 1; (ii) Ignoring item ranks when evaluating recommendation quality can lead to substantial shortcomings. Since recommended items are typically presented in ranked lists, items in higher ranks must be more relevant to keep the user satisfied. Ranking problems differ fundamentally from classification problems, since the position of an item, even within the top- k results, can attract different levels of user attention. Previous work has demonstrated that user attention can drop

Table 1. List of works which use fairness metrics with flaws.

Limitation	Papers
over-simplifying the notion of fairness	[10, 21, 22, 27–30, 45, 46, 56, 61, 66, 69, 82, 84, 85, 93, 97]
not considering rank of items	[10, 21, 22, 29, 45, 61, 69, 82, 84, 85, 93, 97]

sharply after the first few positions in a ranked list [35]. Consequently, rank-aware evaluation provides a more complete way of evaluating models. Some papers that use evaluation strategies that are prone to this issue are included in Table 1. It is important to note however, that even rank-aware metrics can introduce positional bias, which refers to the tendency of users to favor the items that appear on top of a ranked list. Evaluations limited to a fixed cutoff like Recall@k, may lead to a skewed sense of assessment of how the recommendation model performs. Additionally, metrics like MAP (Mean Average Precision), which typically assume binary relevance (0: not relevant and 1: relevant) may fail to capture the nuanced relevance that can come from items having multiple categories. These considerations highlight the importance of using more than one metric (both with and without using ranks) to quantify bias; (iii) relying only on one type of fairness metric can mask underlying biases and create a misleading perception of fairness in recommendation systems. A model that appears fair under one metric can fail to hold other fairness metrics and risk overlooking subtle unfairness issues. Thus, adopting a diverse set of evaluation metrics is essential for quantifying hidden biases and achieving a comprehensive assessment of fairness in RS.

We highlight several studies that have taken into account different classes when evaluating recommendations [33, 47, 48, 53, 59, 60, 73, 74, 76, 87]. For example, [59] groups users based on specific sensitive attributes and categorizes items accordingly, then computes a preference ratio, which is the fraction of items liked by a group in categories. Next, the bias disparity is quantified as the preferences and recommended ratios' relative differences. While this is close to our work it still doesn't account for item rankings, and it does not directly compare recommendations between male and female users, as we do. Additionally, [76] introduces the notion of calibrated recommendations, which aim to align recommended items with user preferences without overemphasizing particular categories. The work by [33] proposes *Skew@k* to evaluate disparities in the proportion recommended candidates across sensitive attributes, and [47] employs Attention Weighted Ranked Fairness (AWRF) [73] to promote balanced exposure in different groups of providers. While both these approaches address group-level fairness, our work differs in its focus on analyzing the distribution of content categories across user groups. Unlike the work by [47] which concentrates on provider-side fairness, we specifically target consumer-side fairness.

3 Proposed Evaluation Metrics

3.1 Motivating Fairness Concern

We consider a typical offline recommendation setting in which the model is trained on historical user–movie interaction data. In this scenario, suppose we have two users: u_1 , a male user who has watched numerous action and sci-fi movies, with some romance movies, and u_2 , a female user who has primarily watched drama movies but has also viewed a few action films. Let us say that we decide to calculate the overall precision separately for the male and female user groups and then compare these values as a means to assess the fairness of the model.

3.1.1 Potential Issue. Machine learning systems are known to learn and amplify biases present in their training data [5, 20, 25, 51, 63]. Recommender systems are no exception, as they can absorb stereotypical patterns from user-item interactions and make biased recommendations based on

sensitive attributes of users [66, 78]. Such biases can have wide-reaching effects on users if the models recommend content just because it conforms to stereotypes, for instance, males like action movies, and females like romance movies.

These imbalanced recommendations may result in users being exposed to items that align with part of their preferences and stereotypical norms. Consequently, diverse content can be filtered out, preventing users from discovering new movies. Over time, u_1 might stop getting romance movies recommended to them, despite having an interest in them. For u_2 , a similar effect may occur with respect to action movies. In essence, such imbalances can confine users to a bubble of recommendations with only their established preferences and gender-stereotypical genres. Additionally, if a user's established preferences already conform to gender stereotypes, then the bias in recommendations will intensify further, giving rise to a filter bubble, leading to increasingly homogeneous and repetitive recommendations.

3.1.2 Falling short when quantifying gender bias. Moreover, as mentioned earlier, using some performance metrics for both genders and comparing them to evaluate fairness is not a great idea. The two groups can have similar scores, even if the model is making biased recommendations by choosing to neglect certain categories for certain users. Our proposed metrics address this issue by breaking down the recommendations by category to get a more nuanced sense of fairness.

The key takeaway here is the importance of learning fair user and item representations, as well as evaluating fairness in the output. Even if a model is trained only on user-item-rating interactions with no explicit mention of sensitive attributes, the model can still infer this private information due to the correlation between their behavior and their sensitive attributes [12, 14, 31]. Therefore, precautionary measures must be taken during training to prevent the model from learning these correlations. We also want to discuss how it is important to ensure personalization, including gender-specific preferences to enhance user satisfaction, but such preferences should be balanced against any risk of reinforcing stereotypes.

Please note that in our study we focus on binary genders, while acknowledging that there are many other gender identities not represented here.

3.2 Notation

We define the notation used for all fairness assessment metrics as follows:

- u_i : An individual user, where i indexes the users.
- v_j : An individual item, where j indexes the items.
- $\mathcal{U}$ and $\mathcal{V}$: The set of users and items, respectively.
- $\mathcal{U}_m$ and $\mathcal{U}_f$: The set of male and female users, respectively.
- c : An item category (e.g., Action, Sci-Fi, Romance)
- C : A category indicator matrix where $C_{j,c} = 1$ if item v_j belongs to category c , and $C_{j,c} = 0$ otherwise.
- C_{v_j} : The set of categories associated with item v_j .
- $TopK_{u_i}$: The set of top K items recommended to user u_i .
- C : Represents the set of all item categories.

To evaluate fairness, we adapt and extend Information Retrieval metrics, proposing both non-ranking and ranking-based measures, which are described in detail in the following subsections.

3.3 Non-ranking-based metrics

The first group of metrics we introduce assesses the fairness of a recommender system without considering the ranking of items.

3.3.1 Category Coverage (CC). This metric measures the proportion of recommended items belonging to category c relative to all categories for all users u_i . In other words, it captures the category-specific distribution of recommended items. Formally, it is defined as follows:

$$CC(c, \mathcal{U}) = \frac{1}{|\mathcal{U}|} \sum_{u_i \in \mathcal{U}} \frac{1}{|TopK_{u_i}|} \sum_{v_j \in TopK_{u_i}} \frac{C_{j,c}}{|C_{v_j}|} \quad (1)$$

3.3.2 Relative Category Representation (RCR). This metric quantifies the proportion of recommended items belonging to category c relative to the overall distributions of all categories from the whole dataset. It provides insights into category-specific items and is defined as follows:

$$RCR(c, \mathcal{U}) = \frac{1}{|\mathcal{U}|} \sum_{u_i \in \mathcal{U}} \frac{\sum_{v_j \in TopK_{u_i}} \frac{C_{j,c}}{|C_{v_j}|}}{\sum_{v_k \in \mathcal{V}} \frac{C_{k,c}}{|C_{v_k}|}} \quad (2)$$

Together, these two metrics enable us to assess how relevant a category is for items recommended to a given set of users. For example, computing CC across different categories c would help us quantify the diversity of items recommended by reflecting the overall distribution of recommended content. In contrast, RCR (when evaluated for different categories c) indicates whether the recommendation system suppresses or amplifies certain categories by comparing the recommended distribution to the true distribution of available content. Both CC and RCR offer a more nuanced view of recommendation behavior, but they don't take into account the position of the recommended items. In many recommendation settings, items are presented sequentially, and the items at the top of the menu must be the most relevant. Consequently, it is essential to consider ranking positions for evaluating recommended items. To address this, in the next section, we introduce rank-based metrics.

3.4 Rank-based Metrics

We now present our rank-aware metrics that incorporate item ordering to provide a comprehensive assessment of fairness.

3.4.1 Category Mean Average Precision (CMAP). For a given user group, this metric estimates the proportion of recommended items belonging to category c while explicitly accounting for their ranking positions. A category-specific average precision score is computed for each user and then averaged across all users. This metric is formally defined as:

$$CMAP(c, \mathcal{U}) = \frac{1}{|\mathcal{U}|} \sum_{u_i \in \mathcal{U}} \frac{\sum_{j=1}^{|TopK_{u_i}|} P(j) \cdot \frac{C_{j,c}}{|C_{v_j}|}}{\sum_{v_j \in \mathcal{V}} \frac{C_{j,c}}{|C_{v_j}|}} \quad (3)$$

where $P(j) = \frac{1}{j} \sum_{k=1}^j \frac{C_{k,c}}{|C_{v_k}|}$.

3.4.2 Category Discounted Cumulative Gain (CDCG). The score for this metric is discounted according to the position of an item in the recommended list. Similar to the previous metric, CDCG uses a logarithmic discount factor, assigning lower weights to items appearing at lower ranks. The metric is defined as follows:

$$CDCG(c, \mathcal{U}) = \frac{1}{|\mathcal{U}|} \sum_{u_i \in \mathcal{U}} \frac{1}{|TopK_{u_i}|} \sum_{j=1}^{|TopK_{u_i}|} \frac{\frac{C_{j,c}}{|C_{v_j}|}}{\log(j+1)} \quad (4)$$

3.4.3 Category Mean Reciprocal Rank (CMRR). This metric is derived from MRR, which computes the mean reciprocal of the rank of the first relevant item. We adapt this metric for the purpose of fairness assessment, and define it as follows:

$$CMRR(c, \mathcal{U}) = \frac{1}{|\mathcal{U}|} \sum_{u_i \in \mathcal{U}} \frac{1}{|TopK_{u_i}|} \sum_{j=1}^{|TopK_{u_i}|} \frac{\frac{C_{j,c}}{|C_{v_j}|}}{j} \quad (5)$$

3.4.4 Category RPrecision (CRP). This metric computes category-specific precision but considering only the top $\lfloor R_c \rfloor$ recommended items, where R_c represents the proportion of category c within the entire item set $\mathcal{V}$ (i.e., $R_c = \sum_{j=1}^{|\mathcal{V}|} \frac{C_{j,c}}{|C_{v_j}|}$, note that item v_j can belong to multiple categories like movie genres). This metric is defined as follows:

$$CRP(c, \mathcal{U}) = \frac{1}{|\mathcal{U}|} \sum_{u_i \in \mathcal{U}} \frac{\sum_{j=1}^{\lfloor Top \lfloor R_c \rfloor_{u_i} \rfloor} \frac{C_{j,c}}{|C_{v_j}|}}{\lfloor R_c \rfloor} \quad (6)$$

All four of the rank-based metrics account for category proportions while incorporating item rank as a discounting factor. Each metric employs a distinct discounting strategy, which captures different aspects of the recommendation behavior. For instance, *CRP* evaluates recommendations only up to a cutoff rank $\lfloor R_c \rfloor$ assessing whether the model allocates a proportionate amount of recommendations for c relative to its presence in the dataset. In contrast, a metric such as *CDCG*, would capture a different dimension of relevance by assigning greater weight to items belonging to category c that appear higher in the ranked list.

3.5 Group Fairness (Gender)

The Gender Balance Score (GBS) for a given category c , is defined as the absolute difference between the category distribution values for $\mathcal{U}_m$ and $\mathcal{U}_f$ calculated using any of the category-aware metrics discussed above. This metric is an adaptation of *demographic parity* as discussed in Section 2.2. Its goal is to ensure that the category distributions of the recommended items for the groups, of males and females are similar. For each category-aware metric $\mathcal{M}$, defined above, we aim for the summed differences between the values calculated for male and female users across all categories to be close to 0. This can be formally defined as:

$$\Delta \mathcal{M}_c = |\mathcal{M}(c, \mathcal{U}_m) - \mathcal{M}(c, \mathcal{U}_f)| \quad (7)$$

$$GBS(\mathcal{M}) = \sum_{c \in \mathcal{C}} \Delta \mathcal{M}_c \approx 0 \quad (8)$$

4 Genre Aware Regularization For Gender Fairness

Having introduced our set of metrics for capturing a nuanced sense of fairness for recommended items, we now focus on learning fair representations that are category-aware for users of different genders. To encourage fairness in recommendation models we adapt two in-processing regularization techniques inspired by the approach first proposed by [95]. We propose to incorporate Category Coverage (Section 3.3.1) and Relative Category Representation (Section 3.3.2) into the loss function alongside the primary recommendation loss, guiding the model to optimize for category-aware fairness. We want to emphasize that while we select *CC* and *RCR* here, any of the other proposed metrics can be incorporated as long as the implementation is differentiable.

Although both regularization terms aim to reduce category-aware gender bias, they focus on different aspects of fairness. The *CC*-based regularization term minimizes disparities in absolute category distributions between male and female overall, encouraging both groups to receive a

similar proportion of recommendations for each category. Essentially, it promotes intergroup parity in exposure to recommendations. In contrast, the RCR-based regularizer term normalizes the category proportions for each group with respect to the underlying distribution of items in the dataset. This helps preserve the natural distribution within the item catalog and prevents the model from over- or under-emphasizing specific categories while reducing demographic bias. To explain the difference between these two fairness schemes clearly, we use an example. If 5% of the movies in a dataset are from the genre Western, CC-based regularizer would aim to minimize the absolute difference in category proportions regardless of the magnitude. But RCR-based regularizer will ensure Western movies are not over-represented (more than 5%) while also focusing on fair exposure for the different groups of users.

This fairness regularizer is applied into the loss function of each baseline model—MF, VAE-CF, LightGCN, XSimGCL and NeuMF.

Specifically, to prevent the model from learning biased representations, we introduce the following regularizer terms:

$$\mathcal{L}_{FairCC} = GBS(CC) = \sum_{c \in C} |CC(c, \mathcal{U}_m) - CC(c, \mathcal{U}_f)| \quad (9)$$

$$\mathcal{L}_{FairRCR} = GBS(RCR) = \sum_{c \in C} |RCR(c, \mathcal{U}_m) - RCR(c, \mathcal{U}_f)| \quad (10)$$

The objective is to minimize disparities in the category distributions of items recommended to users of each sensitive group. The new loss function for MF, VAE-CF, LightGCN, XSimGCL and NeuCF models, incorporating the first regularization term, is defined as follows:

$$\mathcal{L} = \alpha \mathcal{L}_{FairCC} + (1 - \alpha) \mathcal{L}_{Recommendation} \quad (11)$$

The updated loss function for MF, VAE-CF, LightGCN, XSimGCL and NeuCF models, which employ the second regularization term is defined as:

$$\mathcal{L} = \alpha \mathcal{L}_{FairRCR} + (1 - \alpha) \mathcal{L}_{Recommendation} \quad (12)$$

where α controls the trade-off between the primary recommendation loss and the fairness regularization term. $\mathcal{L}_{Recommendation}$ denotes the model-specific recommendation loss. By tuning this hyperparameter, we ensure that fairness constraints do not unduly compromise the recommender system's ability to respect users' personal preferences (even when natural differences in category preferences exist across genders). To regulate the magnitude of the gender-based loss and ensure that it contributes meaningfully to the overall objective, we apply a sigmoid transformation, centering at 0.5 and scaled by 0.1. In addition, the maximum batch loss of the base recommendation objective is used to scale the gender loss up to ensure both losses are on the same scale. In the following section, we demonstrate how the fairness regularization terms help effectively reduce bias in recommendations across different categories for users of different genders.

5 Experiments

In the subsection that follows, we detail the experimental setup used in our study.

5.1 Datasets

Our experiments are conducted on three recommendation datasets which are summarized in Table 2. Each dataset is divided into training, validation, and test sets with a 70:10:20 ratio following a user-based split scheme. It's important to note that items may belong to multiple categories. For clarity and interpretability, we focus on four representative categories when visualizing our metric values—*Action*, *Romance*, *Sci-Fi*, and *Drama* from the MovieLens datasets and *Coffee*, *Tea & Desserts*, *Arts & Entertainment*, *Travel & Transportation* and *Asian* from the Yelp dataset; however

all categories are included in our experimental evaluation. For all of our datasets, we filter out inactive users, defining a user as inactive if they have fewer than five interactions. Additionally, we remove any user whose gender information is unknown. The Yelp dataset has over 300 fine-grained categories. For improved interpretability, we consolidate these into 21 broader categories. These categories include *Active Life & Fitness*, *Arts & Entertainment*, *Automotive*, *Bars & Nightlife*, *Coffee, Tea & Desserts*, *Drinks & Spirits*, *Education & Learning*, *Event Services*, *Family & Kids*, *Food & Restaurants*, *Health & Beauty*, *Home & Garden*, *Miscellaneous*, *Outdoor Activities*, *Public Services & Community*, *Shopping & Fashion*, *Specialty Food & Groceries*, *Sports & Recreation*, *Technology & Electronics*, *Travel & Transportation*, and *Asian*.

5.2 Evaluating Performance

To evaluate recommendation performance, we use HitRatio@ k calculated for each user and then averaged. In addition, we employ a ranking-based metric NDCG@ k (Normalized Discounted Cumulative Gain) which assigns greater relevance to items appearing higher in the ranked list. This metric is also calculated per user using their top k recommendations and then averaged. All performance metrics are reported for $k = 50$.

5.3 Base Recommendation Models

To analyze which models exhibit gender bias, we evaluate several recommendation approaches. Our selection spans both traditional and more recent algorithms. A detailed overview of the model is presented below:

- MF [54]: A classical Matrix Factorization approach in which users and items are represented as latent vectors with global bias. In our implementation, we optimize the model using BPR [70] loss with negative sampling to improve recommendation quality.
- NeuMF [43]: NeuMF is a deep-learning based extension of MF, that integrates the linear structure of traditional MF models and the non-linearity of DNNs (Deep Neural Networks).
- VAE-CF [57]: This non-linear probabilistic model is built on a variational autoencoder architecture and learns compressed information about data. The encoder projects user interactions into a low-dimensional latent space, and the decoder decodes this information back to the high-dimensional vector which is used to make predictions.
- LightGCN [42]: A graph-based collaborative filtering model that simplifies Graph Convolutional Networks (GCN) by retaining only the neighbourhood aggregation operation while removing feature transformation and non-linear activation layers.
- XSimGCL [94]: A graph contrastive learning recommendation model that improves collaborative filtering by learning robust user and item representations through self-supervised contrastive learning. Unlike previous graph contrastive methods that rely on computationally expensive graph augmentations, XSimGCL generates contrastive views by applying simple noise-based embedding perturbations.

We employ the Cornac framework [72], with several custom modifications to support our experimental requirements, including the implementations of XSimGCL and differentiable ways to obtain top- k category distributions required to calculate the gender-based regularization terms across

Table 2. Description of datasets used for our experiments.

Dataset	Users	Item	Interactions	Categories for Items
ML 100K [41]	943	1,349	99,287	18
ML 1M [41]	6,040	3,416	999,611	18
Yelp [64]	1,316	1,272	97,991	21

Table 3. Performance and bias Evaluation results (GBSs) for five baseline models, and their fair counterparts. The fair models here use the *GBS(CC)* as the fairness regularizer.

Model		Hit Ratio↑	NDCG↑	CC↓	RCR↓	CMAF↓	CDCG↓	CMRR↓	CRP↓
ML 100k Dataset									
MF	Original	0.9118±0.0109	0.2034±0.0034	0.0276±0.0021	0.0311±0.0011	0.0039±0.0002	0.0080±0.0006	0.0046±0.0007	0.0241±0.0020
	Fair	0.8931±0.0079	0.1810±0.0021	0.0176±0.0047	0.0207±0.0058	0.0026±0.0008	0.0049±0.0015	0.0029±0.0012	0.0152±0.0014
VAE-CF	Original	0.8810±0.0132	0.1539±0.0033	0.0125±0.0038	0.0139±0.0054	0.0012±0.0004	0.0027±0.0006	0.0018±0.0005	0.0073±0.0038
	Fair	0.8676±0.0126	0.1547±0.0023	0.0072±0.0037	0.0090±0.0032	0.0008±0.0002	0.0019±0.0005	0.0012±0.0004	0.0050±0.0014
NEU-MF	Original	0.9432±0.0022	0.2487±0.0022	0.1637±0.0045	0.1335±0.0060	0.0221±0.0004	0.0450±0.0011	0.0178±0.0008	0.1276±0.0055
	Fair	0.9149±0.0076	0.2106±0.0028	0.0377±0.0056	0.0408±0.0060	0.0066±0.0008	0.0121±0.0014	0.0062±0.0006	0.0364±0.0057
LightGCN	Original	0.8927±0.0061	0.2025 ±0.0020	0.0364±0.0012	0.0400±0.0012	0.0054±0.0001	0.0100±0.0003	0.0040±0.0002	0.0378±0.0005
	Fair	0.8676±0.0084	0.1927±0.00245	0.0375±0.0045	0.0396±0.0037	0.0058±0.0008	0.0106±0.0011	0.0045±0.0008	0.0378±0.0026
XSimGCL	Original	0.8026±0.0063	0.1205±0.0019	0.1352±0.0053	0.1099±0.0031	0.0237±0.0009	0.0381±0.0016	0.0153±0.0008	0.1167±0.0041
	Fair	0.8134±0.0052	0.1265±0.0011	0.1438±0.0050	0.1174±0.0032	0.0254±0.0010	0.0408±0.0018	0.0167±0.0013	0.1272±0.0048
ML 1M Dataset									
MF	Original	0.8452±0.0030	0.1515±0.0011	0.1786±0.0065	0.0488±0.0015	0.0098±0.0003	0.0500±0.0014	0.0208±0.0004	0.1445±0.0030
	Fair	0.8086±0.0113	0.1156±0.0037	0.1364±0.0166	0.0398±0.0064	0.0076±0.0010	0.0386±0.0048	0.0165±0.0028	0.0764±0.0077
VAE-CF	Original	0.8047±0.00674	0.1306±0.0022	0.0378±0.0607	0.0116±0.0167	0.0021±0.0038	0.0102±0.0181	0.0043±0.0078	0.0216±0.0357
	Fair	0.8034±0.0066	0.1307±0.0016	0.0157±0.0111	0.0054±0.0040	0.0007±0.0006	0.0039±0.0033	0.0018±0.0015	0.0094±0.0074
NEU-MF	Original	0.9194±0.0022	0.1533±0.0022	0.3163±0.0039	0.1030±0.0012	0.0250±0.0005	0.0848±0.0011	0.0317±0.0006	0.2460±0.0025
	Fair	0.8126±0.0077	0.1103±0.0412	0.1103±0.0412	0.0332±0.0124	0.0065±0.0029	0.0315±0.0125	0.0135±0.0056	0.0708±0.0115
LightGCN	Original	0.9111±0.0084	0.1777±0.0015	0.3507±0.0063	0.1141±0.0008	0.0260±0.0013	0.0939±0.0020	0.0350±0.0011	0.2849±0.0049
	Fair	0.8637±0.0095	0.1479±0.0013	0.3138±0.0183	0.1093±0.0063	0.0263±0.0011	0.0840±0.0048	0.0314±0.0017	0.2508±0.0154
XSimGCL	Original	0.9035±0.0024	0.1663±0.0010	0.3102±0.0016	0.1069±0.0010	0.0308±0.0002	0.0825±0.0004	0.0304±0.0001	0.2533±0.0016
	Fair	0.8169±0.0034	0.1111±0.0019	0.2739±0.0018	0.0975±0.0011	0.0310±0.0011	0.0736±0.0004	0.0276±0.0001	0.1617±0.0051
Yelp Dataset									
MF	Original	0.8663±0.0269	0.1374±0.0390	0.0349±0.0035	0.0362±0.0034	0.0049±0.0007	0.0086±0.0010	0.0034±0.0003	0.0284±0.0012
	Fair	0.8028±0.0623	0.1095±0.0537	0.0236±0.0138	0.0282±0.0101	0.0032±0.0012	0.0060±0.0027	0.0023±0.0007	0.0185±0.0103
VAE-CF	Original	0.8041±0.0083	0.0966±0.0015	0.0027±0.0005	0.0029±0.0005	0.0006±0.0002	0.0008±0.0003	0.0006±0.0004	0.0029±0.0007
	Fair	0.7926±0.0134	0.0925±0.0019	0.0021±0.0010	0.0027±0.0021	0.0003±0.0002	0.0004±0.0001	0.0003±0.0001	0.0016±0.0005
NEU-MF	Original	0.9354±0.0040	0.1762±0.0028	0.0509±0.0061	0.0839±0.0035	0.0131±0.0013	0.0140±0.0019	0.0058±0.0009	0.0516±0.0050
	Fair	0.8020±0.01452	0.0932±0.0034	0.0036±0.0023	0.0057±0.0041	0.0017±0.0009	0.0019±0.0011	0.0017±0.0010	0.0073±0.0034
LightGCN	Original	0.8944±0.0069	0.2032±0.0018	0.0580±0.0038	0.0896±0.0086	0.0147±0.0014	0.0160±0.0012	0.0066±0.0006	0.0505±0.0022
	Fair	0.8615±0.0023	0.1921±0.0004	0.0533±0.0051	0.0827±0.0074	0.0142±0.0022	0.0150±0.0017	0.0065±0.0009	0.0504±0.0051
XSimGCL	Original	0.8746±0.0110	0.1401±0.0018	0.0575±0.0031	0.1035±0.0074	0.0159±0.0012	0.0166±0.0004	0.0073±0.0007	0.0566±0.0023
	Fair	0.8319±0.0027	0.1141±0.0017	0.0541±0.0040	0.0995±0.0098	0.0136±0.0016	0.0157±0.0008	0.0070±0.0006	0.0526±0.0047

different models. When computing our regularization terms, we generate a top- k recommendation list with $k = 50$. Moreover, all Gender Balance Score (GBS) results (except CRP) use the top 50 recommended items. For GBS(CRP) we use top $\lfloor R_j \rfloor$ as defined in Equation 6.

5.4 Experimental Settings

To improve robustness and statistical rigor of our experimental evaluations, all experiments, including stochastic components are repeated using five independent random seeds. For each evaluation metric (including performance and bias), we report the mean performance together with standard deviation (mean±standard deviation) across the five runs. In addition, we report 95% confidence intervals (refer to Figure 10, Figure 11 and Figure 12) to quantify the uncertainty associated with the estimated mean scores.

6 Analysis of the proposed metrics

In this section, we present and analyze the results of our experiments.

6.1 Baselines Comparison

6.1.1 Bias Evaluation. We present our results in Figures 2, 3, 4, 5, 6 and Tables 3, 4. Among the graph-based models, the results don't reflect that one of them is uniformly more gender-biased than the other, which is also observed by [7]. For the chosen biased categories, XSimGCL tend to show more bias than LightGCN (as shown in Figure 2, however, when the scores are aggregated for the fairness metrics (as shown on Tables 3 and 4), LightGCN shows higher scores. In general, since LightGCN just propagates and aggregates neighbour embeddings via simplified linear GCN layers, it is more susceptible to manifesting bias when compared to XSimGCL. In contrast, XSimGCL augments this process with an InfoNCE-based contrastive learning objective that promotes embedding uniformity

Table 4. Performance and bias Evaluation results (GBSs) for five baseline models, and their fair counterparts. The fair models here use the *GBS(RCR)* as the fairness regularizer.

Model		Hit Ratio↑	NDCG↑	CC↓	RCR↓	CMAF↓	CDCG↓	CMRR↓	CRP↓
ML 100k Dataset									
MF	Original	0.9118±0.0109	0.2034±0.0034	0.0276±0.0021	0.0311±0.0011	0.0039±0.0002	0.0080±0.0006	0.0046±0.0007	0.0241±0.0020
	Fair	0.8681±0.0075	0.1586±0.0030	0.0247±0.0057	0.0301±0.0064	0.0034±0.0006	0.0070±0.0018	0.0039±0.0010	0.0199±0.0028
VAE-CF	Original	0.8764±0.0121	0.1530±0.0028	0.0125±0.0038	0.0139±0.0054	0.0012±0.0004	0.0027±0.0006	0.0018±0.0005	0.0073±0.0038
	Fair	0.8744±0.0097	0.1535±0.0021	0.0092±0.0023	0.0091±0.0025	0.0010±0.0002	0.0023±0.0004	0.0015±0.0005	0.0057±0.0015
NEU-MF	Original	0.9375±0.0119	0.2297±0.0429	0.1637±0.0045	0.1335±0.0060	0.0221±0.0004	0.0450±0.0011	0.0178±0.0008	0.1276±0.0055
	Fair	0.9025±0.0119	0.02008±0.0257	0.0393±0.0076	0.0416±0.0056	0.0070±0.0012	0.0120±0.0019	0.0060±0.0007	0.0355±0.0028
LightGCN	Original	0.8927±0.0061	0.2025±0.0020	0.0364±0.0012	0.0400±0.0012	0.0054±0.0001	0.0100±0.0003	0.0040±0.0002	0.0378±0.0005
	Fair	0.8676±0.0084	0.1927±0.0025	0.0375±0.0045	0.0398±0.0028	0.0058±0.0008	0.0106±0.0009	0.0045±0.0008	0.0370±0.0027
XSimGCL	Original	0.8030±0.0071	0.1206±0.0021	0.1353±0.0055	0.1099±0.0031	0.0237±0.0009	0.0381±0.0016	0.0153±0.0008	0.1168±0.0041
	Fair	0.8121±0.0072	0.1266±0.0020	0.1452±0.0054	0.1187±0.0048	0.0258±0.0010	0.0412±0.0019	0.0169±0.0013	0.1274±0.0044
ML 1M Dataset									
MF	Original	0.8452±0.0030	0.1515±0.0011	0.1786±0.0065	0.0488±0.0015	0.0098±0.0003	0.0500±0.0014	0.0208±0.0004	0.1445±0.0030
	Fair	0.8016±0.0157	0.1147±0.0081	0.1410±0.0168	0.0412±0.0078	0.0078±0.0015	0.0389±0.0047	0.0159±0.0029	0.0763±0.0129
VAE-CF	Original	0.8047±0.0067	0.1306±0.0022	0.0378±0.0607	0.0116±0.0167	0.0021±0.0038	0.0102±0.0181	0.0043±0.0078	0.0216±0.0357
	Fair	0.8029±0.0056	0.1306±0.0009	0.0163±0.0147	0.0057±0.0045	0.0008±0.0007	0.0041±0.0041	0.0017±0.0016	0.0099±0.0085
NEU-MF	Original	0.9194±0.0021	0.1534±0.0020	0.3163±0.0039	0.1030±0.0012	0.0250±0.0005	0.0848±0.0011	0.0317±0.0006	0.2460±0.0025
	Fair	0.8072±0.0078	0.1283±0.0030	0.1227±0.0393	0.0372±0.0108	0.0075±0.0026	0.0332±0.0100	0.0129±0.0038	0.0747±0.0122
LightGCN	Original	0.9076±0.0056	0.1773±0.0011	0.3529±0.0047	0.1135±0.0006	0.0254±0.0009	0.0947±0.0014	0.0354±0.0007	0.2870±0.0026
	Fair	0.8021±0.0051	0.1488±0.0013	0.2848±0.0069	0.0965±0.0022	0.0233±0.0007	0.0765±0.0013	0.0288±0.0003	0.2239±0.0054
XSimGCL	Original	0.9037±0.0027	0.1663±0.0010	0.3103±0.0016	0.1070±0.0009	0.0308±0.0002	0.0825±0.0004	0.0304±0.0001	0.2534±0.0016
	Fair	0.8142±0.0043	0.1106±0.0016	0.2733±0.0035	0.0976±0.0020	0.0309±0.0015	0.0738±0.0007	0.0279±0.0002	0.1600±0.0053
Yelp Dataset									
MF	Original	0.8526±0.0053	0.1190±0.0024	0.0349±0.0035	0.0362±0.0034	0.0049±0.0007	0.0086±0.0010	0.0034±0.0003	0.0284±0.0012
	Fair	0.7918±0.0116	0.0856±0.0044	0.0196±0.0030	0.0232±0.0042	0.0023±0.0006	0.0049±0.0010	0.0019±0.0004	0.0147±0.0024
VAE-CF	Original	0.8041±0.0083	0.0966±0.00150	0.0027±0.0005	0.0029±0.0005	0.0006±0.0002	0.0008±0.0003	0.0006±0.0004	0.0029±0.0007
	Fair	0.7894±0.0128	0.0924±0.0017	0.0021±0.0007	0.0028±0.0019	0.0003±0.0002	0.0005±0.0001	0.0003±0.0001	0.0019±0.0005
NEU-MF	Original	0.9354±0.0041	0.1762±0.0028	0.0509±0.0061	0.0839±0.0035	0.0131±0.0013	0.0140±0.0019	0.0058±0.0009	0.0516±0.0050
	Fair	0.8046±0.0028	0.0931±0.0025	0.0036±0.0032	0.0057±0.0056	0.0017±0.0013	0.0019±0.0014	0.0016±0.0011	0.0080±0.0051
LightGCN	Original	0.9618±0.0042	0.2026±0.0021	0.0580±0.0038	0.0896±0.0086	0.0147±0.0014	0.0160±0.0012	0.0066±0.0006	0.0505±0.0022
	Fair	0.9510±0.0036	0.1898±0.0021	0.0543±0.0049	0.0807±0.0084	0.0146±0.0020	0.0152±0.0016	0.0066±0.0010	0.0510±0.0040
XSimGCL	Original	0.8745±0.0101	0.1402±0.0020	0.0574±0.0031	0.1043±0.0071	0.0159±0.0013	0.0165±0.0004	0.0072±0.0004	0.0568±0.0025
	Fair	0.8469±0.0089	0.1143±0.0024	0.0545±0.0040	0.1012±0.0091	0.0141±0.0013	0.0157±0.0008	0.0067±0.0003	0.0551±0.0037

and representation robustness. This regularisation may discourage over-specialised representations that encode demographic-specific interaction patterns, thereby mitigating the propagation of gender bias. Among all the models, the graph-based models are more prone to producing biased recommendations. This has been observed in previous studies as well [4, 17]. It is due to graph propagation which has been identified as a potential mechanism of bias amplification in graph-based recommender systems [90].

For the other baseline models, the MF-based approaches generally have higher bias scores. In MF models, we have explicit embeddings to encode user and item representations, which can encapsulate correlations between user behaviours and their demographic attributes (like gender). VAE-CF, which uses variational autoencoders to learn probabilistic latent representations of users, is less prone to producing highly biased recommendations. NeuMF has an architecture that combines Generalized Matrix Factorization (GMF) (linear) and Multi-Layer Perceptrons (MLP) (non-linear) components. This helps capture complex user-item interactions. Meaning this combination can also capture stereotypical patterns from the training process. This results in stronger bias in the latent representations. Given this complexity, NeuMF's higher bias scores compared to the MF model are consistent with our expectations. For smaller datasets (ML-100k and Yelp), the VAE-CF model shows relatively lower bias scores. This is likely due to limited interactions that prevent the model from fully learning biased preferences. In contrast, for the 1M dataset, the same model is seen to manifest higher bias scores, meaning it was able to capture the stereotypical patterns. Across all models, higher gender bias scores are obtained on the larger dataset than the smaller ones.

All baseline models yield *GBS* scores $\neq 0$, indicating that there is discriminatory recommendations where certain categories are exposed to users of certain gender. As shown on Tables 3 and 4, *GBS* values are higher for *CC* and *RCR* (ref to Equations: 1 and 2). Both of which are non-ranked metrics and do not employ any discounting based on where an item appears on the recommendation list.

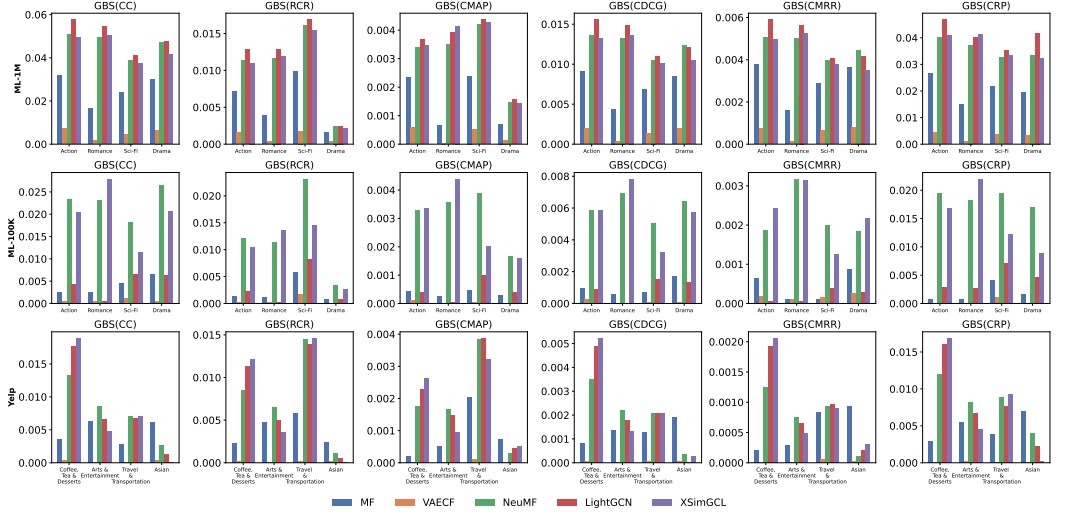


Fig. 2. Comparison of gender bias values measured using the six metrics we propose, for all three datasets.

In contrast, *GBS(CMRR)* shows the lowest values for all models, since it has a strong discounting for lower-ranked items.

After applying each regularization term, we observe a significant reduction in bias scores, especially for the NeuMF model, as illustrated in Figures 3, 4 and Tables 3, 4. The effect of bias mitigation is most pronounced for NeuMF model, where *GBS(CC)* achieves an average bias reductions of 71%, 66.4%, and 86% (across all 6 metrics) on the ML-100K, ML-1M, and Yelp datasets, respectively, while *GBS(RCR)* yields reductions of 70.8%, 64.1%, and 86.1% on the same datasets. Overall, *GBS(CC)* shows a slightly higher reduction in bias scores, although the results of using them as regularization terms are largely comparable (for an in-depth analysis, please refer to Section A.2). The effect of bias reduction is less pronounced for LightGCN. The fairness terms penalise group disparity in the recommendations. However, it does not directly change which neighbours are aggregated or how much biased information is transmitted through each edge. From the heatmaps, we can observe that among the four stereotypical genres considered, *Action* and *Romance* exhibit higher bias scores, with action movies being recommended more to male users and romance to female users. In the Yelp dataset, notable biases are present for the categories: *Coffee, Tea & Desserts* and *Travel & Transportation*. Where the former is more often recommended to female users and the latter to male users.

6.1.2 Performance Evaluation. The performance degradation observed can be considered minimal. However, the extent of this drop is highly dependent on the choice of the hyperparameter α (see Section 6.1.3). For all models, α , is selected to balance the trade-off between bias and performance, ensuring we reduce bias while avoiding a substantial loss in quality of recommendations. In certain cases, the fair variant can outperform the base model. Although this is counterintuitive, it can be attributed to the fairness term acting as a regularizer that effectively mitigates overfitting. This phenomenon has also been observed in previous works [45, 50, 53].

6.1.3 Impact of the hyperparameter α . In this section, we analyze the impact of α on both recommendation performance and fairness. As described in Section 4, α controls the relative strength of

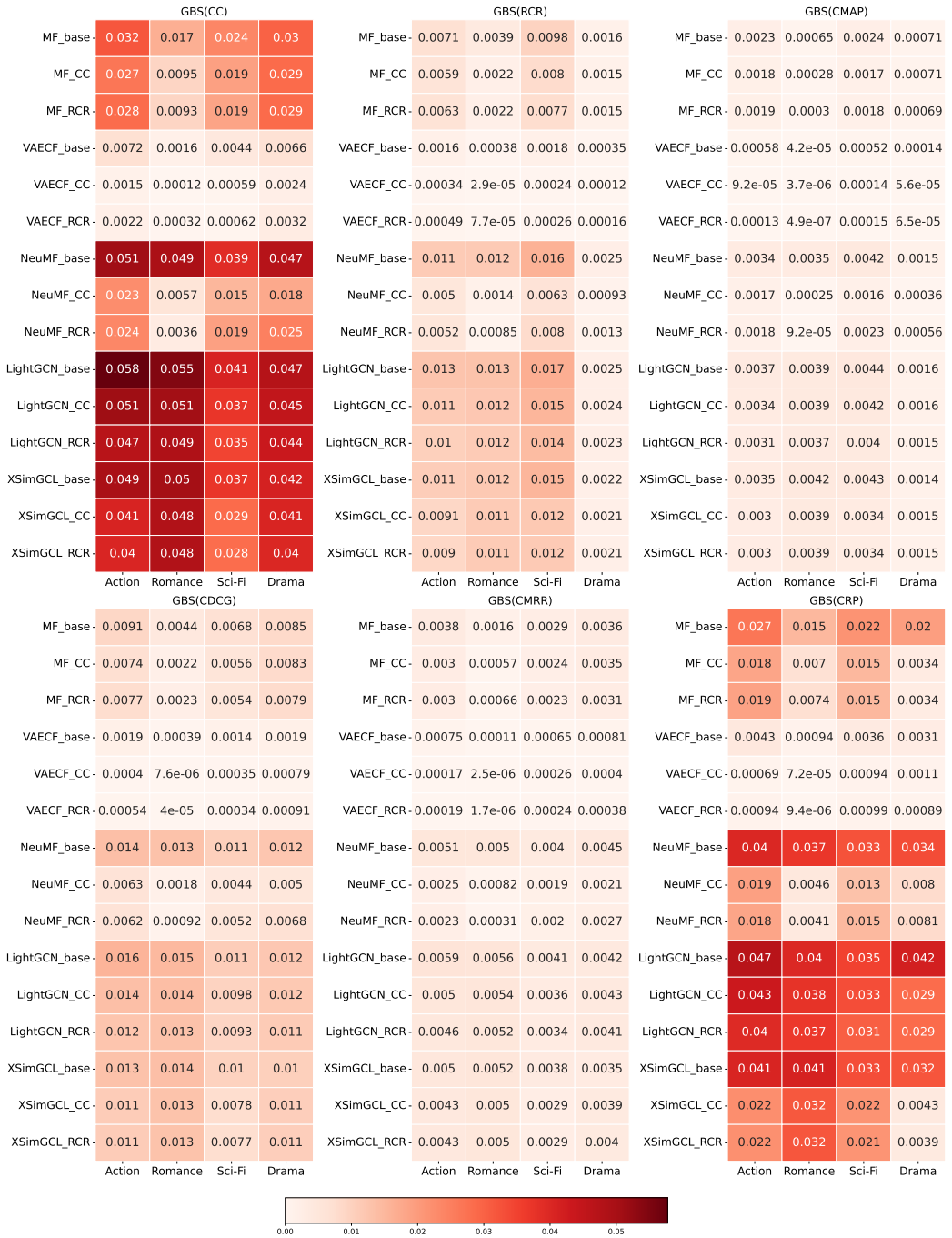


Fig. 3. Heatmap of bias scores (calculated using all 6 metrics) for baseline models and their fairness-aware variants, using both regularization terms. The visualization is across four stereotypical genres in the ML-1M dataset.

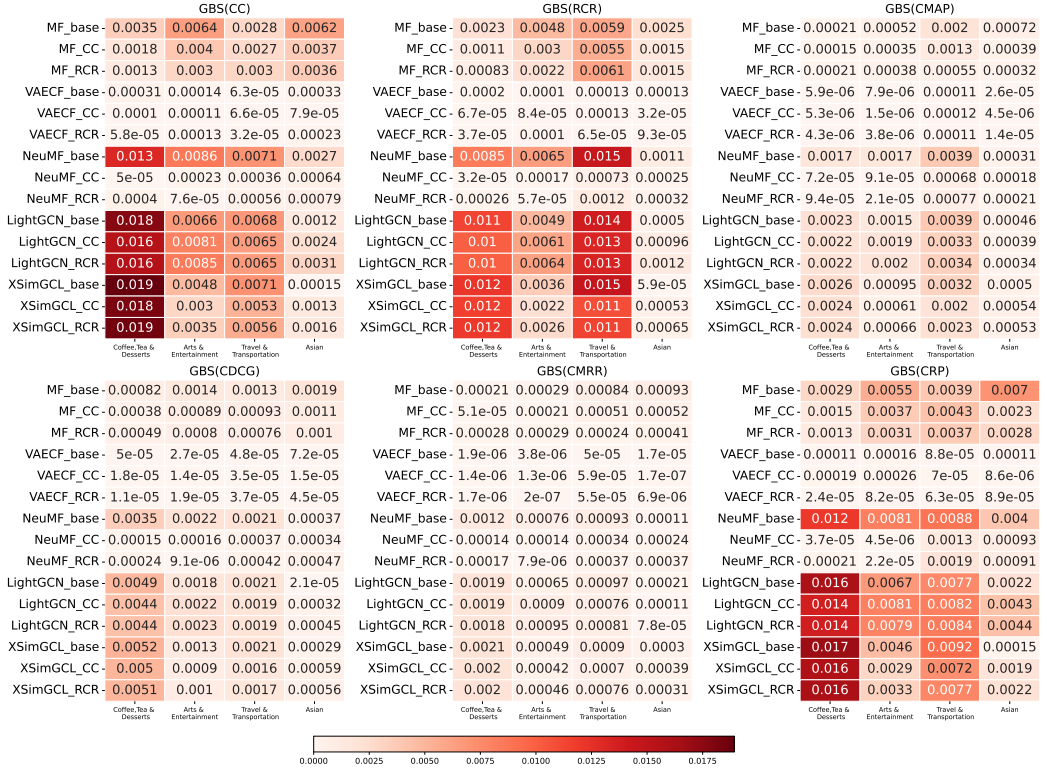


Fig. 4. Heatmap of bias scores (calculated using all 6 metrics) for baseline models and their fairness-aware variants, using both regularization terms. The visualization is across four stereotypical categories in the Yelp dataset.

the fair regularizer term. In theory, increasing this would lead to a decrease in recommendation performance, as the weight of the primary recommendation loss is reduced.

We use different α values ranging from 0 to 0.6 in increments of 0.1 for the experiments. Values greater than 0.6 are not used, since that would mean overpowering the loss function with the fairness term. This would essentially undermine the main objective of accurate recommendations. To examine the trade-off between fairness and performance, we plot the optimized fairness metric for each regularizer term along with a recommendation performance metric (in our case, we used NDCG). As shown on Figures 5 and 6, both performance and bias generally decrease as α increases. Some fluctuations are observed, especially when the value of α is low, where the bias initially increases before declining. We hypothesize this is due to the fairness term being too weak at smaller values of α to meaningfully reduce bias, while still perturbing the main optimization process. As α grows larger and the fairness term contributes more to the loss function and there is a consistent reduction in bias. For both the regularization terms, the bias drop is comparable as mentioned in Section 6.1.1. This overall pattern is also observed for the graph-based models, indicating that the fairness term remains effective in most settings. However, LightGCN and XSimGCL exhibit occasional non-monotonic fluctuations across particular values of α and individual bias metrics. Unlike MF, VAEF and NeuMF, LightGCN and XSimGCL repeatedly propagate information through the user-item interaction graph, allowing structural biases to persist during message passing. Since

the proposed fairness regulariser operates only at the optimisation objective and does not modify the graph propagation process, increasing α does not necessarily produce a monotonic reduction across all bias metrics.

Additionally, in some cases, there is a slight improvement in performance even when α is increasing. As mentioned before, this can be explained by the fairness term acting as a regularizer to prevent overfitting. We select α values by ensuring there is bias reduction with minimal performance loss. For more details on what alpha was selected, please refer to Table 5.

7 Conclusion

In this paper, we identify the key limitations of current metrics used to quantify consumer-side bias in recommendation systems. To address this issue, we proposed a set of six metrics to accurately quantify gender bias in recommendation models. We do this by incorporating the item categories in our metrics that allow a more nuanced and fine-grained assessment of fairness in the recommended items.

We further demonstrate that integrating two of these metrics as fairness-aware regularization terms along with the main recommendation loss can effectively reduce category-level biases related to gender, with minimal performance loss. We perform extensive experiments on three real-world datasets across a diverse set of recommendation models, and show that our metrics can effectively

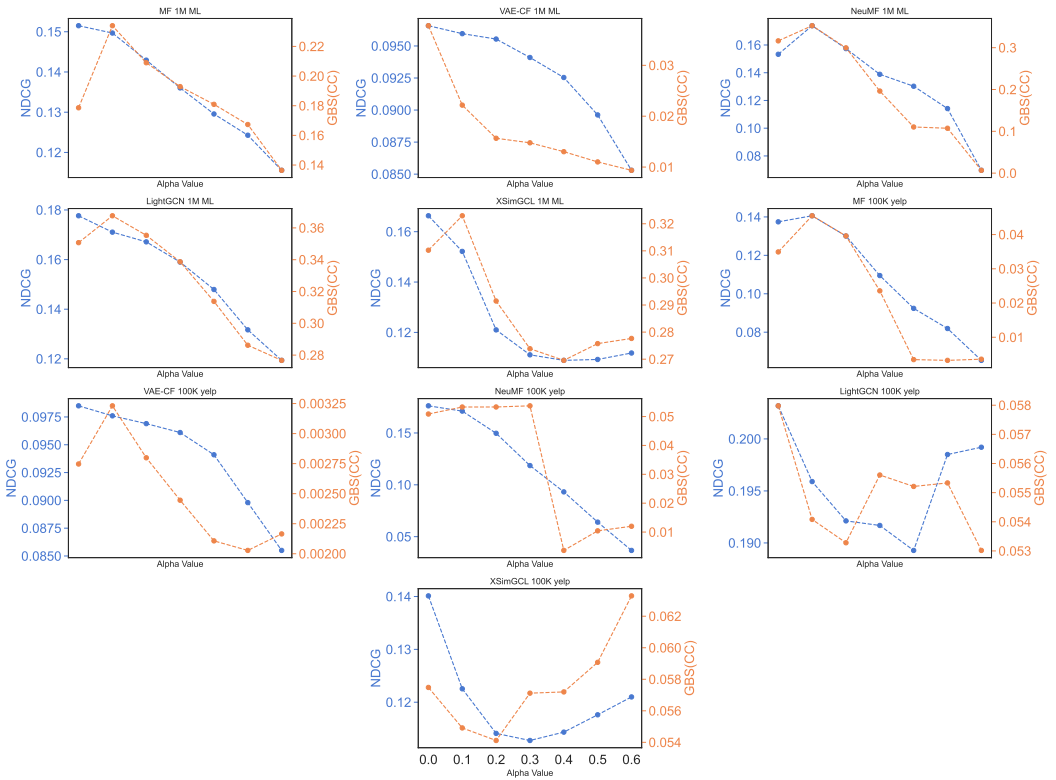


Fig. 5. Effect of the regularization weight α on recommendation performance (NDCG@50) and the bias measure, which is the difference of Category Coverage values for male and female users (GBS(CC)), with experiments conducted on all three datasets.

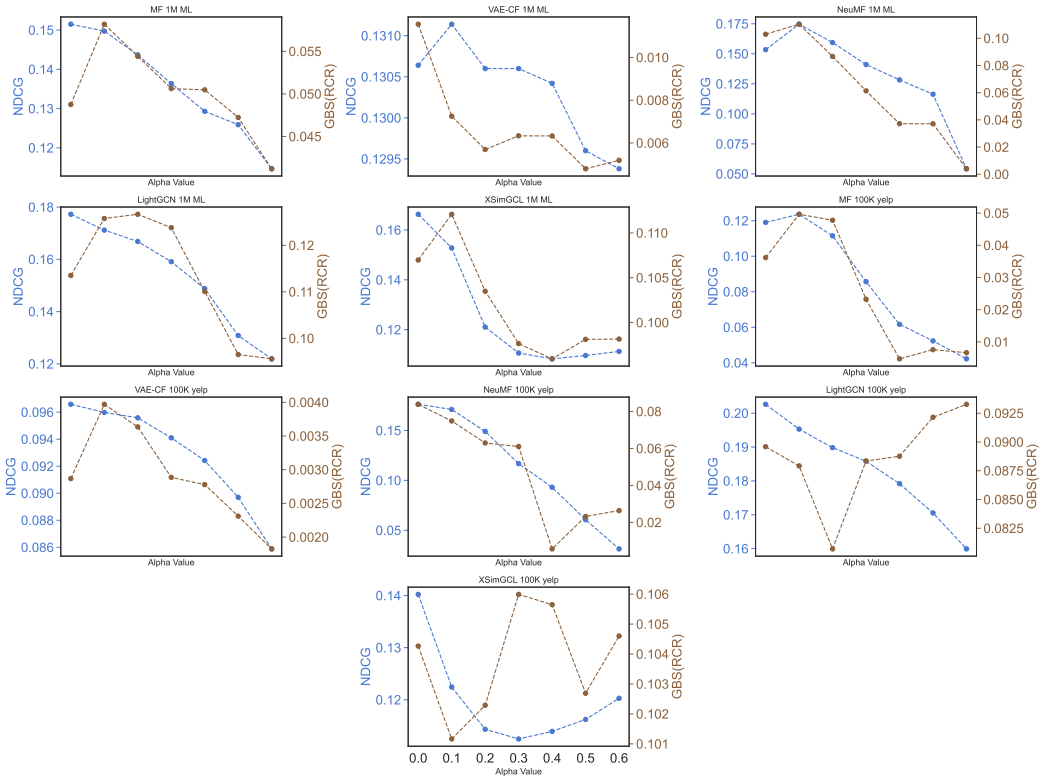


Fig. 6. Effect of the regularization weight α on recommendation performance (NDCG@50) and the bias measure, which is the difference of Relative Category Representation values for male and female users (GBS(RCR)), with experiments conducted on all three datasets.

uncover biases in a granular way. After introducing the regularization term, the models exhibit a substantial reduction in bias while maintaining a favorable balance between fairness and accuracy.

A key strength of our fairness approach is its flexibility. The proposed fairness-aware loss terms allow for context-sensitive fairness. For instance, in high-stakes domains such as job or loan recommendation systems, the fairness term can be prioritized, whereas in lower-risk domains such as music recommendation, accuracy can be favored. Our work highlights the risk of relying on overly simplistic fairness metrics, which can give practitioners a false sense of equity in recommendations. We argue that using a more granular approach like the one proposed here is necessary to understand and mitigate such biases.

Finally, the metrics proposed here are highly versatile and can be extended beyond consumer-side fairness¹. They can be adapted to measure provider-side fairness by using popularity or brands of items. In the future, we plan to extend this to support multi-valued and intersectional sensitive attributes.

Generative AI Disclosure: The paper and the ideas presented herein are written entirely by the

¹It is important to mention that while our work primarily focuses on gender-based consumer fairness, we acknowledge that this framework implicitly aligns with CP-fairness too, since we also consider item characteristics.

Table 5. Alpha values used to weigh the fairness term for both GBS(CC) and GBS(RCR)

		ML-100K	ML-1M	Yelp
NeuMF	GBS(CC)	0.2	0.4	0.4
	GBS(RCR)	0.2	0.4	0.4
VAECF	GBS(CC)	0.5	0.2	0.4
	GBS(RCR)	0.3	0.2	0.4
MF	GBS(CC)	0.6	0.6	0.3
	GBS(RCR)	0.5	0.6	0.3
LightGCN	GBS(CC)	0.2	0.4	0.2
	GBS(RCR)	0.2	0.5	0.2
XSimGCL	GBS(CC)	0.1	0.3	0.2
	GBS(RCR)	0.1	0.3	0.1

authors. ChatGPT and Claude were used to paraphrase selected portions of the text to improve clarity and readability. The original ideas and conclusions are all from the authors.

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A Appendix

This section presents additional experimental results and details that complement the findings reported in the main paper.

A.1 Process of generating Figure 1

This plot is based on the ML-100K dataset. After training the models, we identify the users by their gender in the test set. To ensure a fair comparison, we use the number of users in the smaller group, which was female in our case. Next, we randomly chose the same number of male users from the test set. Once we have an equal number of male and female users, Precision@10 is then computed for each user and averaged separately for each gender. To determine the genre distribution of movies recommended, we generate the top 10 movies for each user. The proportion of each genre is calculated using Equation 1.

A.2 Comparison of the two regularizer terms

The results after employing the two different regularizer terms are highly comparable, as shown in Figure 13. This makes sense since the two terms are nearly equivalent, the only difference being in the normalization. As discussed before, theoretically, using RCR should preserve relative category distribution better than using CC. Since the results shown in the tables are those of the aggregated category values, we use Figure 14, for a more in-depth analysis of using the different regularizer terms. From the figure, we can see that RCR-based regularizer was consistently able to produce recommendations with category distribution closer to the underlying distributions of the actual dataset when compared to the CC-based regularizer. Overall, both regularizers are effective at reducing category-aware gender bias, while the RCR-based formulation offers a modest advantage when preserving the relative prevalence of categories in the underlying item catalogue is important. It is important to note that, the objective of this work is not to establish one regularization strategy as universally superior to the other. Rather, our goal is to demonstrate that category-aware fairness can be effectively incorporated into recommendation models using different regularization objectives while providing a more granular evaluation of recommendation bias. The two regularizers emphasize slightly different notions of fairness, and the choice between them should ultimately depend on the requirements of the application.

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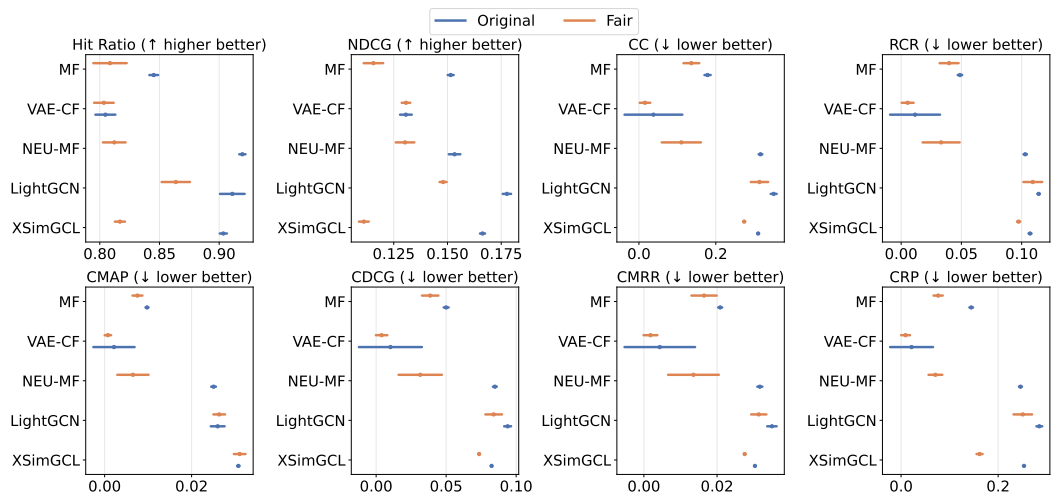


Fig. 7. 95% confidence intervals for the performance and bias evaluation results for five baseline models and their fair counterparts based on CC. The results are for the dataset: ML1M

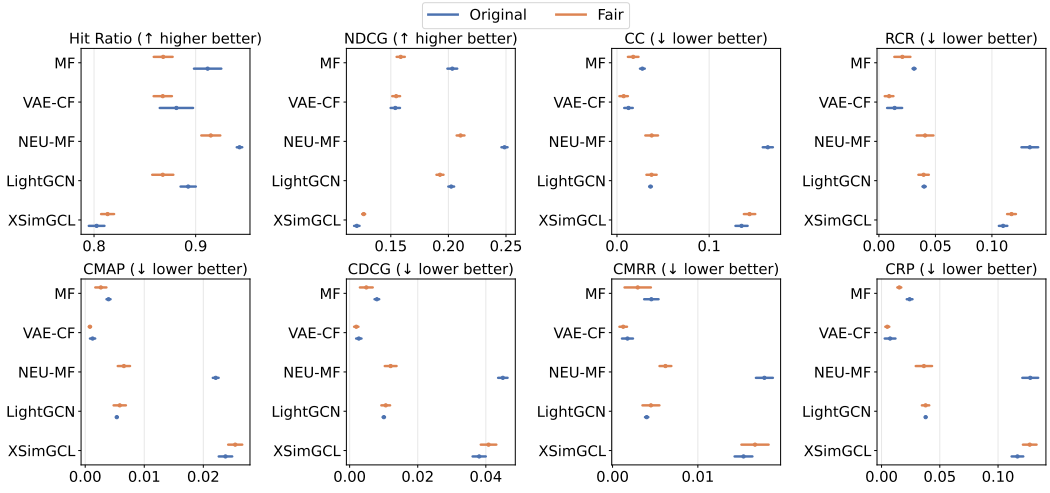


Fig. 8. 95% confidence intervals for the performance and bias evaluation results for five baseline models and their fair counterparts based on CC. The results are for the dataset: ML100K

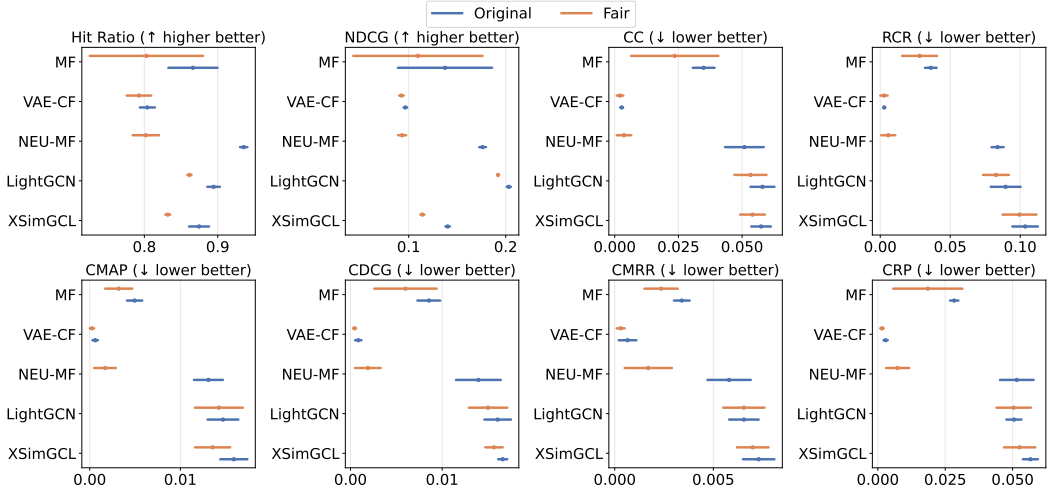


Fig. 9. 95% confidence intervals for the performance and bias evaluation results for five baseline models and their fair counterparts based on CC. The results are for the dataset: Yelp100k

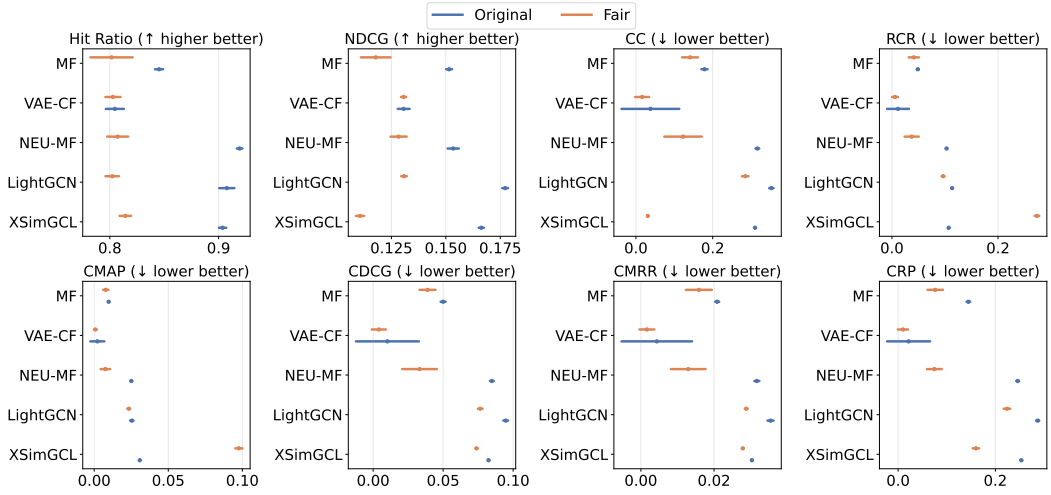


Fig. 10. 95% confidence intervals for the performance and bias evaluation results for five baseline models and their fair counterparts based on RCR. The results are for the dataset: ML1M

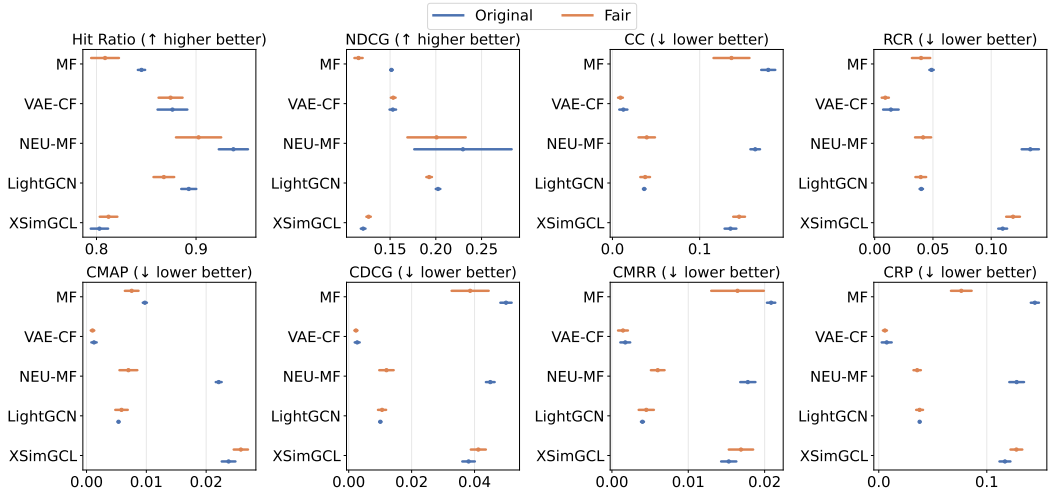


Fig. 11. 95% confidence intervals for the performance and bias evaluation results for five baseline models and their fair counterparts based on RCR. The results are for the dataset: ML100K

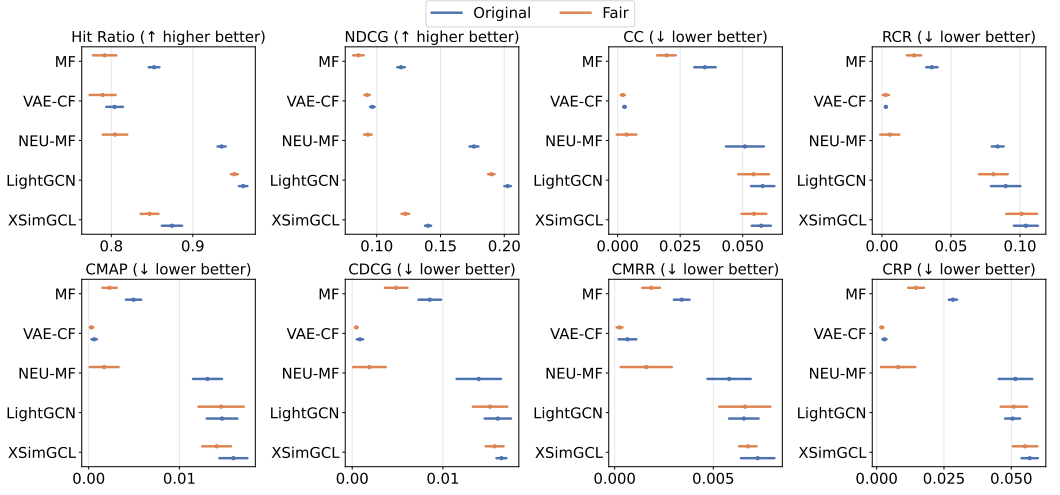


Fig. 12. 95% confidence intervals for the performance and bias evaluation results for five baseline models and their fair counterparts based on RCR. The results are for the dataset: Yelp100k

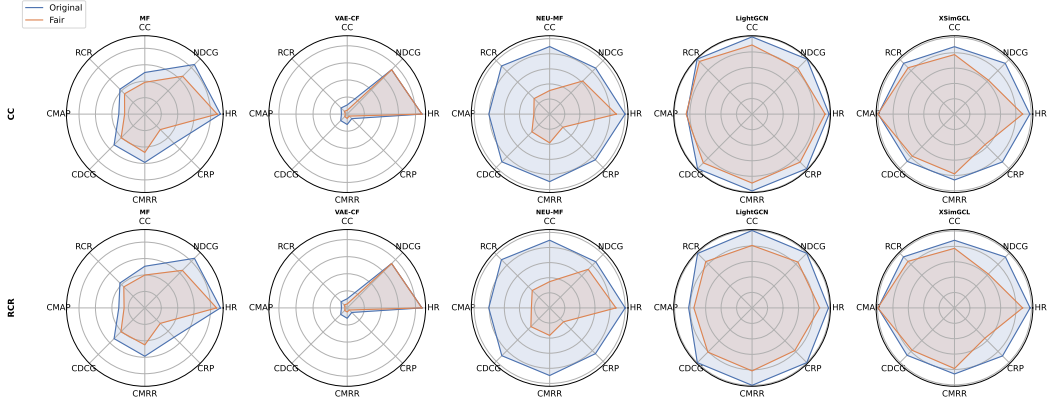


Fig. 13. Fairness CC vs RCR for Original vs Fair Profile per Model (axes normalized 0-1 across both regularizers for direct comparability)

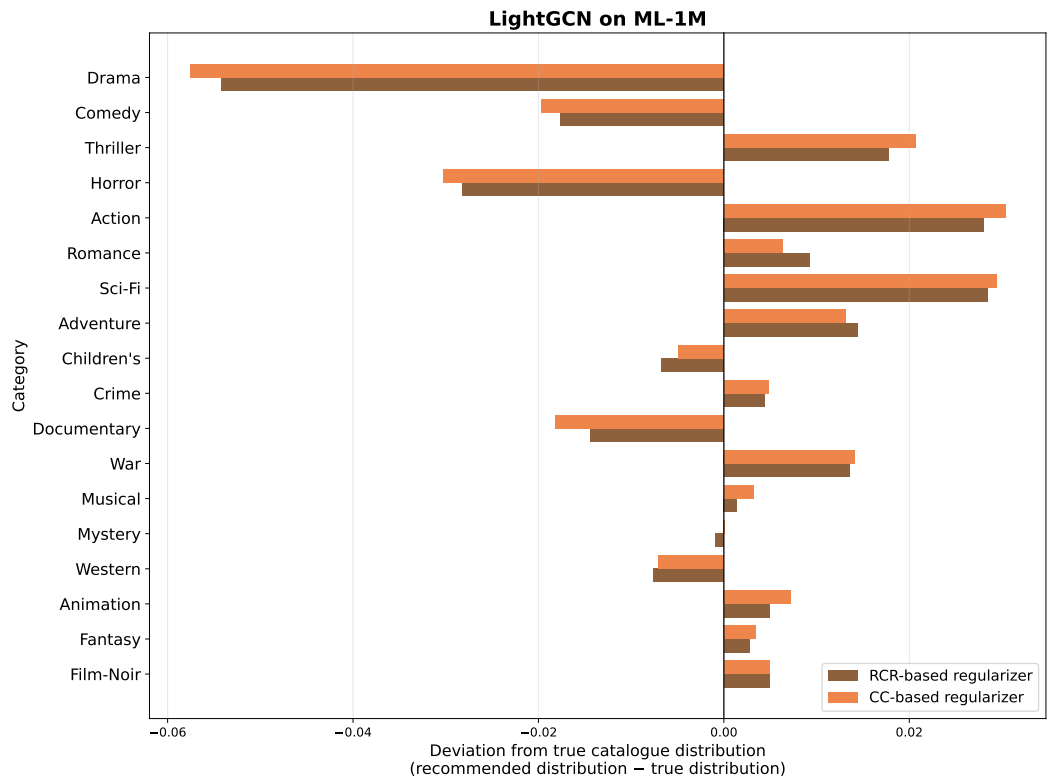


Fig. 14. Deviation from true catalog distribution, by model: RCR-based vs CC-based regularizer. The results are displayed for the best-performing model, LightGCN (in terms of both HR and NDCG) for ML1M.